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ORIGINAL RESEARCH

A Robust Modified Character Segmentation Approach for the Handwritten Archaic Modi Documents

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Abstract

Character segmentation is an imperative step of intelligent handwritten archaic Modi document recognition system. Numbers of challenges like inconsistent and non-uniform handwritten characters, broken or degraded characters etc. are faced in the process of character segmentation due to age, cursive and stylish writing nature of Modi script. This paper presents a modified preliminary segmentation step for segmentation of Modi intact characters and overlapping/touching characters cluster to overcome problem of under/bad segmentation. Here, a zone based three steps Modi script character segmentation approach is presented. At preliminary step, column wise background and foreground pixels are scrutinized to determine leading segmentation of text line. Modi text line is segmented in two types of segments as isolated characters and cluster of overlapping/touching characters. Local zoning based method and column wise background pixel density exploration is used to segment overlapping and touching characters clusters respectively. Performance of the proposed method and comparative analysis is verified using MODI-HHDoc Modi document dataset. Successful Modi character segmentation rate is achieved as 87.70%. As compared to previous hybrid background pixel density based technique, bad segmentation rate is reduced from 0.8 to 0.2%. Comparative analysis shows that proposed modified Modi character segmentation technique is more efficient to state-of-art benchmarking techniques.

Keywords Modi script · Character segmentation · Zoning · Pixel density

Introduction

The important official, social, cultural, historical, religious and even personal information is found in written form on papers or paper like substances from prehistoric era. These substances are likely to be in destroying state due to number of reasons like aging, chemical preservatives etc. [15, 21] The information written in these substances is important in current era for different purposes like historical and social study, judicial and official matters and so on. Consequently, it is desirable that conversion of hard form to soft form i.e. digitization for long period preservation of these documents. As well, it is desirable for the applications like automated recognition, transcription, transliteration, and information retrieval systems.

The ancient Brahmi based Modi is cursive, stylish and unconstrained script derived from Nagari family [15, 21]. Boru/Tak (Wooden stick) or feather of a bird was used to write the Modi script in early age. A straight line (Shirorekha) is drawn before writing and characters are written without lifting the pen. The words are not separated by space. Figure 1 shows examples of Modi documents written in ancient age. Number of challenges are faced during the implementation of the Modi script character segmentation system as listed in Table 1 and described in Fig. 2. The Challenge No. 1 (Ch1) indicates that text line with non-uniform Shirorekha indicated by blue-line in the Fig. 2. Shirorekha is curved or having a slope. Likewise Challenge No. 2 (Ch2) indicates touching characters and so on.

Higher accuracy of character segmentation is one of the most requisite aspect for achieving improved accuracy in document recognition. Character segmentation approaches broadly categorized as explicit (segmentation based) and implicit (holistic) approaches [1, 2, 5, 6]. Traditional explicit

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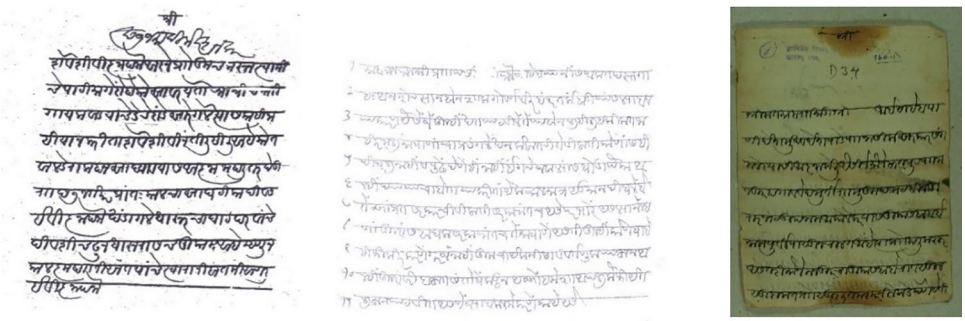


Fig. 1 Examples of original archaic handwritten degraded Modi script documents

Table 1 Modi script character segmentation challenges

Challenge No	Challenges
Ch1	Non-uniform Shirorekha
Ch2	Touching characters
Ch3	Overlapping characters
Ch4	Similar contours
Ch5	Inconsistent character's height and width
Ch6	Conjunct characters
Ch7	Skewed or curved text-line
Ch8	Degraded characters (broken, faded ink etc.)
Ch9	Non-uniform baseline
Ch10	Cursive style writing

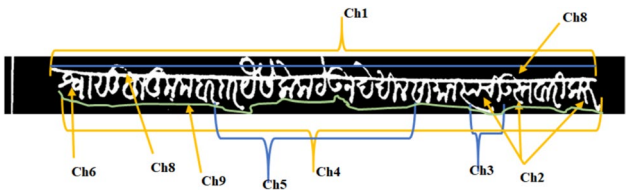


Fig. 2 Illustration of character segmentation challenges listed in Table 1

approaches are failed to find the segmentation point of touching and overlapping text part and may perform over segmentation or under segmentation [5]. In implicit approach intelligent system has been used to estimate touching points in the text. But these approaches are failed to identify overlapping and touching text part. These approaches required huge training data and time that creates extensive overheads [5]. Segmentation approach is extremely relying on writing nature of the script [1]. Hence, character segmentation accuracy for handwritten script is still low.

Here, a novel and modified character segmentation system is proposed to segment Modi characters from Modi text line segment. It is a solution to tackle all types of challenges listed in Table 1. This proposed modified algorithm of preliminary segmentation is based on background pixel intensity exploration. To segment a cluster of overlapping and touching characters two separate methods are devised. Overlapping characters clusters are segmented by employing local text region zoning with column wise background pixel analysis. Likewise, bottom up intensity of background pixel is examined to estimate ligature and segment touching characters.

In this paper, brief introduction about Modi script and research work is given in Sect. Introduction. Benchmarking state-of-art handwritten character segmentation literature is reviewed in Sect. Literature review. Strategy of the proposed Modi character segmentation system is illustrated in Sect. Proposed method. Experimental and comparative results are reported and discussed in Sect. Results and discussions. Finally, conclusion is summarized in Sect. Conclusion.

Literature Review

To obtain high recognition accuracy, several character segmentation approaches are reported in literature which are broadly classified into three kinds as: (i) Explicit Segmentation, (ii) Implicit Segmentation or Recognition Based Segmentation, and (iii) Holistic or segmentation free Approach [1]. In the period of 70's, problems of touching and overlapping unified characters segmentation are identified. Several character segmentation approaches have been reported to deal with touching and overlapping character segmentation challenges for different scripts. However, character segmentation accuracy is still low, and it is very crucial to achieve improvement in touching, overlapping and cursive handwritten characters segmentation rate [5]. Handwritten character segmentation approaches found vary significantly for the handwritten scripts like English, Chinese, Devanagari, Bengali, Guajarati etc. These approaches are script specific and taking number of character segmentation issues into consideration like use of prior knowledge about script's character structuring, writing style of script etc. Specifically, no character segmentation approach is reported for handwritten ancient and cursive Modi script documents.

Number of vertical projection profile based character segmentation approaches are reported in literature for various scripts. Vertical projection profile based method is efficient with respective the non-cursive script and printed text. Modifications are included and basic methodologies are extended as per the requirement of script style. Vertical segmentation algorithm using shape and geometrical knowledge is proposed by [1] to segment handwritten touching English characters. The authors found that technique is not able to eliminate limitations of character segmentation as over segmentation, miss segmentation and bad segmentation. Likewise, vertical projection profile based character segmentation approach is discussed in [24] for multilingual documents. Structural properties are used for the estimation of segmentation columns between characters. Graph distance theory approach is used to separate touched and overlapped characters. Both these approaches are not able to locate accurate boundaries between broken and overlapped characters. Similarly, vertical projection profile based character segmentation approach proposed for different scripts like Brahmi script [25], Urdu script [26], Assamese [28] and Hindi (Devanagari script) [29]. All these are non-cursive script and not efficient to segment the handwritten, cursive characters.

Research work presented in [4, 12] and [17] proposed text based approaches. They used structural properties to segments touching Devanagari characters. These approaches having number of limitations as: (i) Script specific techniques, (ii) Parametric and having over-segmentation problems, (iii) Failed to segment characters with broken left modifier and vertical left modifier; and (iv) Do not tackle with broken characters, overlapping/ touching characters.

Research work proposed in [3] used Self Organizing Feature Maps (SOM) (Region Based) for segmentation of cursive handwritten touching English characters. This method is not efficient to estimate accurate ligature location of touching characters if its compound includes large black pixels. Likewise, the method is not able to segment characters if text contains broken characters.

To segment isolated, overlapping and touching Devanagari characters, Graph distance theory based approach is proposed in [7]. It is a script specific approach and having two inefficiencies. First inefficiency is that post processing is required for locating accurate boundaries between touching characters and second inefficiency is vertical cut estimation between overlapped characters. Similarly, another script specific constrained based Devanagari touching character segmentation system is presented in [14]. Bounding box with vertical bars are analysed to estimate ligature. This approach is failed to segment broken and overlapping characters. An iterative method using connected component with drop flow approach is presented by [23] for the Devanagari ancient manuscripts. In this method characters are segmented without removing headline. Efficiency of this method is depending on two features of character as thickness and size. This method is having over and under segmentation problem. A zone with connected component based character segmentation approach for Bangla and Devanagari script is presented by [10]. The method is script specific and having over segmentation problem.

Morphological operation with minutiae detection algorithm is presented in [13] for segmentation of Devanagari compound characters. Authors are not elaborated that method is related to compound characters means touching or overlapping characters or not. In [11] the authors have discussed header line strengthening and gray value analysis algorithm for the segmentation of Devanagari overlapping characters. These approaches do not efficient with broken part modifiers.

A simplified version of transformation-based learning theory for cursive handwritten touching English character segmentation is discussed in [20]. The main key point of this approach is that training dataset required in character segmentation process.

Character segmentation system for cursive, handwritten, stylish and unconstrained script like Modi script requires more reliable technique which should tackle all the challenges described in Sect. Introduction as listed in Table 1. The research article presented by [14], reported three opinions as: (i) Vertical projection profile-based character segmentation approaches are not suitable for free style, handwritten cursive, unconstrained overlapping characters; (ii) Character segmentation approaches using Hidden Markov Model does not work well with overlapping characters; and (iii) Water reservoir-based techniques are constrained based and are not able to segment overlapping characters [18]. Holistic and recognition based techniques requires annotated training dataset for segmentation. Overall review reveals that most of the approaches are script specific, parametric and constrained or rule based.

Proposed Method

Unconstrained handwritten Modi text line is the input of the proposed Modi character segmentation system. These text lines are segmented from the anciently handwritten Modi documents. Proposed handwritten Modi character segmentation algorithm comprises three main steps as: (i) Preprocessing, (ii) Preliminary segmentation, and (iii) Overlapping/ touching cluster segmentation. Architecture of this system is described in Fig. 3.

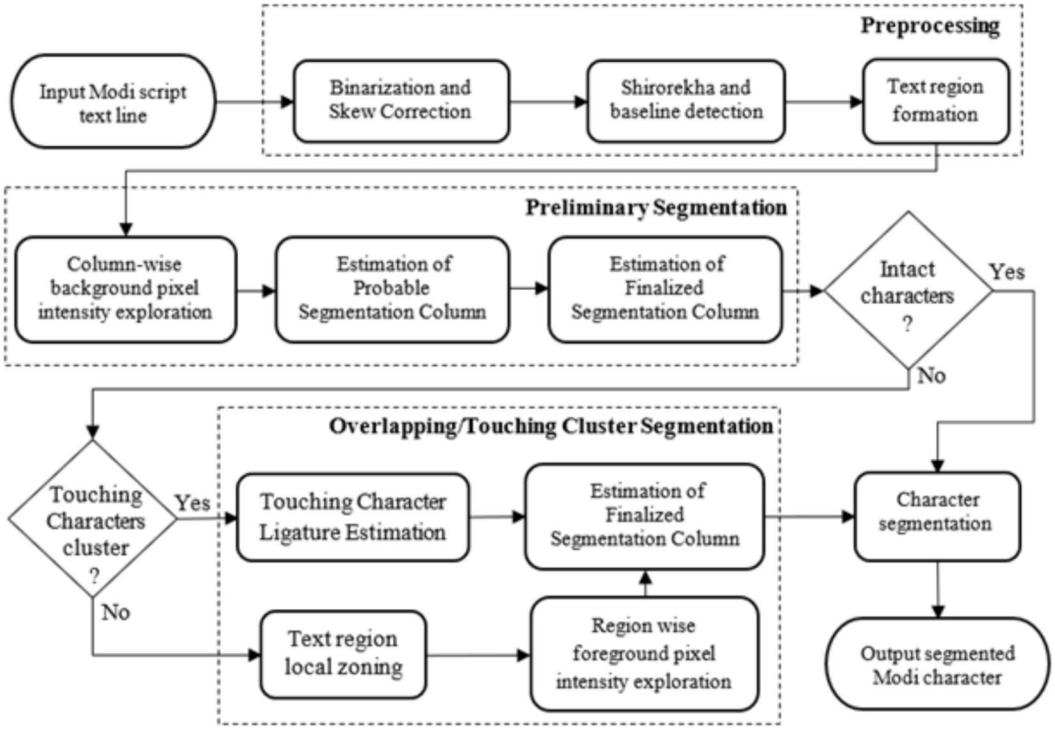
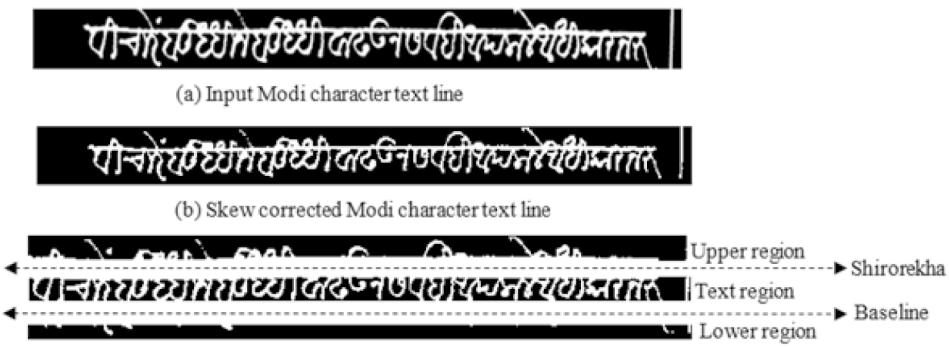


Fig. 3 Modi character segmentation system architecture

Preprocessing

Preprocessing step includes three sub stages as: (i) Text line image binarization and skew correction; (ii) Shirorekha and baseline detection; and (iii) Text region formation. Shirorekha (header line) is drawn by hand before to write a Modi text. Consequently, segmented handwritten Modi text line is having skew. Inputted Modi text line is a gray level image. Gray to binary form conversion is required for the skew correction process. Hence, these text lines are binarized using Otsu's thresholding method [16]. The divide-and-conquer based algorithm presented in [15] is employed for skew detection and correction. Initially, skew angle range of Modi text line is estimated using concentration of the black pixels. To detect skew angle projection profile based criterion function is used. This skew correction approach is correcting skew of Modi text line with 0.1 degree angular resolution [15]. Examples of binarized and skew corrected Modi text lines are shown in Fig. 4b and Fig. 5b. Handwritten Modi text line has two common challenges non-uniform Shirorekha (Table 1: Challenge Ch1) and non-uniform baselines (Table 1: Challenge Ch9). Also, text line may have curved Shirorekha (Table 1: Challenge Ch7). Shirorekha is retained with text line. Consequently, correct estimation of Shirorekha

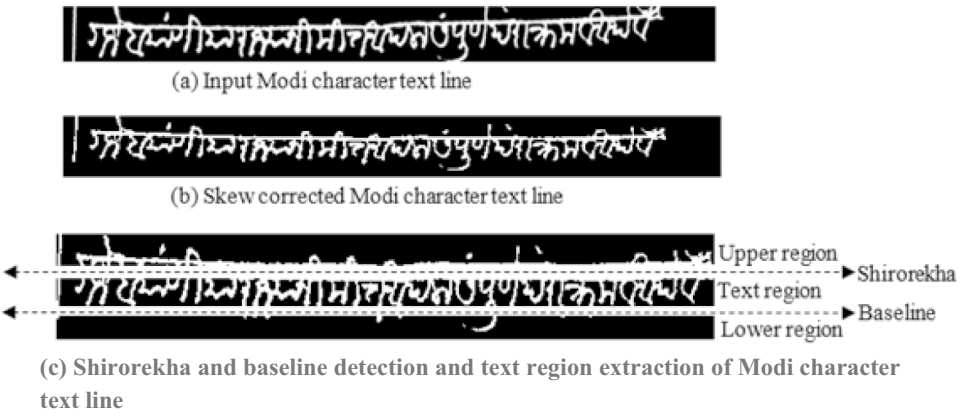


(a) Input Modi character text line

(b) Skew corrected Modi character text line

(c) Shirorekha and baseline detection and text region extraction of Modi character text line

Fig. 4 Step wise illustration of preprocessing of Modi text line



(a) Input Modi character text line

(b) Skew corrected Modi character text line

(c) Shirorekha and baseline detection and text region extraction of Modi character text line

Fig. 5 Step wise illustration of preprocessing of Modi text line with non-uniform Shirorekha

and base line locations are intolerable. Hence, indeterminate location of Shirorekha (SL) and base line (BL) is detected by calculating maximum and minimum peak of horizontal projection profile [8, 22] using Eq. (1) and Eq. (2) respectively. Further, these values are utilized for detection and detachment of upper region, text region and lower region as illustrated in Fig. 4c and Fig. 5c.

$$\mathrm{SL} = \mathrm{maximum}_{(j=1:N)}\left(\sum_{i=1}^{M} I(i,j)\right) \tag{1}$$

$$\mathrm{BL} = \mathrm{minimum}_{(j=1:N)}\left(\sum_{i=1}^{M} I(i,j)\right) \tag{2}$$

Here, I(i, j) is the pixel value at (i, j) location in the text line image I of size M X N. SL is the location of Shirorekha and BL is location of the baseline in text line image I.

Preliminary Segmentation

Intact and clusters of overlapping/touching characters are separated in preliminary segmentation step. Input for this step is text region (MReg) of text line. Each column of text region (MReg) of size H X W is scanned from bottom side. Location of foreground pixel from bottom side for each column is measured. A temporary location (Loc) vector is involved foreground pixel location if it's location is one or one-fourth of height of text region. It is represented mathematically in Eq. (3).

$$Loc = Loc \cup \begin{cases} r & \mathrm{MReg}_{r,c} = 1 \mathrm{A} r \geq {}^{1}/_{4}H \\ \text{none} & \text{otherwise} \end{cases} \quad where,\ r = H : 1\ and\ c = 1 : W \tag{3}$$

Further, non-consecutive locations are calculated by analysing temporary location (Loc) vector. These non-consecutive locations are utilized to estimate Probable Segmentation Column (PSC). Probable Segmentation Column is Nseg X 2 matrix contains location pair (X, Y) of each segmented character region as shown in Fig. 6b. Here, Nseg is the number of segmented characters in text line. These probable segmented regions (SegFig) may contain wrongly segmented region as shown in Fig. 6b and consequently bad segmentation or under segmentation rate is increased. To overcome this problem Finalized Segmentation Column (FSC) is calculated. Foreground pixel intensity (IntSeg) of each probable character region (Segfig) is computed using Eq. (4).

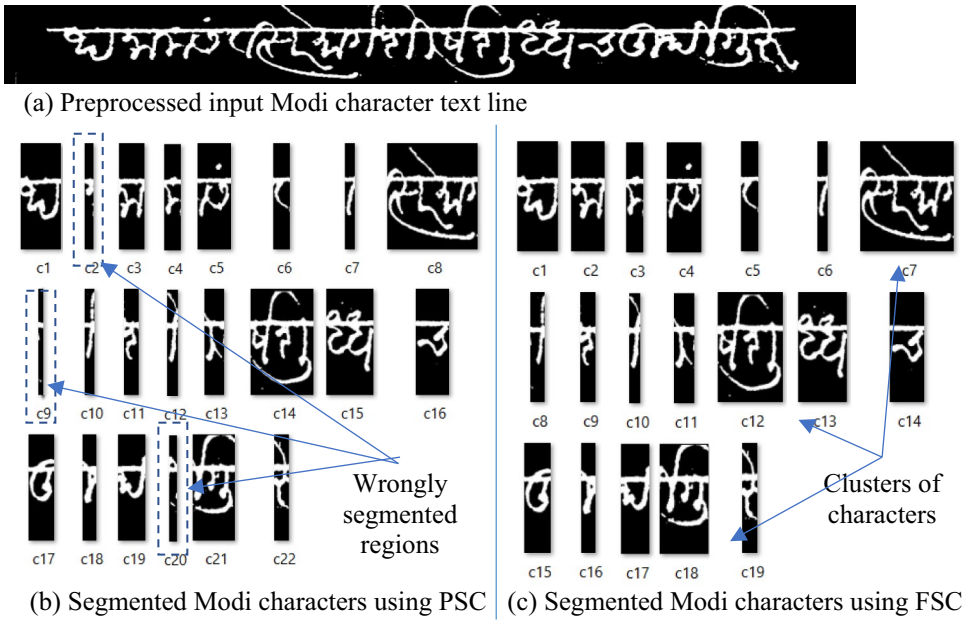
$$IntSeg(c)_{1 \leq c \leq Nseg} = \sum_{i=1}^{H1} \sum_{j=1}^{W1} Segfig(c) \tag{4}$$

Here, IntSeg(c) is foreground pixel intensity of the cth probable character region Segfig(c) of size H1 X W1.

Finalized Segmentation Columns (FSC) are estimated by analyzing intensity of probable character region(IntSeg) and mean intensity of all intensity of all probable character regions (Segfig) using Eq. (5). Figure 6c shows the example of final segmented Modi characters after removal of wrongly segmented character using Finalized Segmentation column.

$$FSC = FSCU \begin{cases} (PSC(i,1), PSC(i+1,2) & IntSeg(i) \leq mean(IntSeg) \\ otherwise & (PSC(i,1), PSC(i,2)) \end{cases} \tag{5}$$

Illustration of preliminary segmentation process of Modi character segmentation system is as shown in Fig. 6 (c1, c2, … are actual file names used to store the segmented characters) and process is algorithmically described in Algorithm 1, Algorithm 2 and Algorithm 3.



(a) Preprocessed input Modi character text line

(b) Segmented Modi characters using PSC

(c) Segmented Modi characters using FSC

Fig. 6 Preliminary segmentation of Modi characters

Algorithm 1: MCharSegment (I)

Input: I – Grayscale text line image.
Output: Schar – Segmented character images

1: I' ⟵ correctSkew(I) // Detect and correct skew of image I
2: I_1 ⟵ img_Binarizarion (I) // Gray to binary image conversion.
3: [Lb , Ls] ⟵ findSB (I_1, H, W) //Find location of Shirorekha and baseline.
4: [Left, Right, SegI] ⟵ findLeftRightEnd(I_1) // Text line image with region of interest.
5: [UReg, MReg, LReg] ⟵ separteUML(segI , Lb, Ls) // Find text zone region MReg
6: [SegLoc] ⟵ preSegment(MReg)
7: Fec ⟵ postProcessing(segLoc ,SegI)
8: Schar ⟵ segCharacter(Fec, SegI)
9: return

Algorithm 2: PreSegment (MReg)

Input: MReg – Middle text region
Output: SegLoc – Estimated segmentation columns

1: [H, W] ⟵ heightWidthOfImage (MReg) // Find height and width of image
2: for c ⟵ each column Mreg
for r ⟵ each column in MReg from bottom side with step -1
if isForeground(MReg(r,c)) // Check foreground pixel/background pixel
if r ≥ ¼ × H
Loc ⟵ Loc U c
endIf
endIf
endFor
3: for l ⟵ each value of Loc // Calculate estimated character region locations
if Loc(l) and Loc(l+1) are not consecutive
SegLoc ⟵ SegLoc U Loc(l)
endIf
endFor
4: return

Algorithm 3: PostProcessing (SegLoc, SegI)

Input: SegLoc – Estimated segmentation columns
SegI – Text line image with region of interest.
Output: Fsc – Finalized segmentation columns

1: for l ⟵ each value of SegLoc from second location // Create segmentation location
// sequence with segLoc
Seq ⟵ Seq U SegLoc(l)
endFor
2: for i ⟵ each value ine segLoc and Seq // Calculate foreground pixel intensity
Psc ⟵ Psc U (SegLoc(i), Seq(i)) // Compute the Probable segmentation
//column pairs
IntSeg ⟵ IntSeg U pixelInt(SegI(Psc(i)) // Compute the foreground pixel
//intensity of each predictable segment
endFor
3: i ⟵ 1 // initialize character counter
4: while i < size(Psc) // Compute Finalized segmentation column pairs
if IntSeg(i) ≤ mean(IntSeg)
Fsc ⟵ Fsc U (Psc(i,1), Psc(i+1, 2))
i⟵ i+2
else
Fsc ⟵ Fsc U (Psc(i,1), Psc(i, 2))
i⟵ i+1
endIf
endWhile
if i ≠ size(Psc)
Fsc=Fsc U (Psc(size(Psc),1), Psc(size(Psc),2));
endIf
5: return

Character Cluster Segmentation

Unsegmented characters means cluster of touching and overlapping characters as shown in Fig. 6c are segmented using two-step process. Location of the Shirorekha and baseline is detected by calculating the maxima and minima of the horizontal projection profile using Eq. (1) and Eq. (2) respectively and text region of the cluster is separated. This text region of cluster is process through two steps:

Step I: Overlapping Character Cluster Segmentation

Overlapping text region is partitioned into four equal zones. Column-wise background pixel density of second and third zone is calculated. These values are analysed and segmentation column location is estimated [22].

Step II: Touching Character Cluster Segmentation

Ligature or segmentation column location for touching characters cluster is found by analysing bottom-up histogram of background pixels. The column wise histogram is calculated up to the location of the foreground pixel come across from the bottom side. These values are analysed and segmentation column location is estimated [22].

Finally, Crucial Segmentation Column (CSC) is identified to segment intact Modi characters by scrutinising the segmentation column locations. This full process of Modi touching and overlapping cluster segmentation is presented in [22]

Results and Discussions

Performance of the proposed character segmentation method is evaluated using evaluation strategy presented in [19] and [22]. It uses five evaluation factors Correct Segmentation Rate (CSR), Successful Segmentation Rate (SSR), precision, recall and F-measure given in Eqs. 6, 7, 8, 9, 10 respectively. These aspects are figure out by counting the number of matches between resultant segmented characters by the proposed appraoch and manually calculated ground truth characters in text line segment [22].

$$CSR = \frac{NC_R - \left(NC_C + NC_B\right)}{NC_G} \times 100 \tag{6}$$

$$SSR = \frac{NC_1}{NC_R} \times 100 \tag{7}$$

$$\text{Precision} = \frac{NC_I + NC_C}{NC_R} \tag{8}$$

$$\text{Recall} = \frac{NC_I + NC_C}{NC_G} \tag{9}$$

$$\text{FM} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{10}$$

Here,

CSR: Correct Segmentation Rate.

SSR: Successful Segmentation Rate.

NC_G: Number of ground truth characters.

NC_R: Number of resultant characters.

NC_I: Number of correctly segmented isolated characters.

NC_C: Number of overlapping or touching characters clusters.

NC_B: Number of incorrectly segmented characters.

To evaluate performance of the proposed system, archaic original handwritten MODI-HHDoc Modi document dataset [32, 33] is used. Three thousand handwritten archaic Modi documents have been collected from various archaeological sectors like Shri Rajwade and Shri Samarth Wagh Devta Sanshodhan Mandal, Dhule, Maharashtra, India etc. this Modi document dataset contained both scanned images by scanner and images captured by camera. These document images are in different size and resolution. These images have numbers of degradations as shown in Fig. 1. Thus, initially these document images are processed by three steps: (i) Frame detection and removal approach presented in [31], (ii) Noise detection and removal approach presented in [16] and (iii) Skew detection and correction approach presented in [15]. Further, these pre-processed document images are segmented in the text lines. Hybrid text line segmentation technique presented in [21] is used for the segmentation of Modi text lines. These segmented text lines are given over to the proposed character segmentation system. Since, ground truth values are not available for testing the performance of this system. Thus, 33,314 characters comprised in 1072 text lines of ninety three Modi documents are manually calculated and ground truth values are evaluated. Modi character segmentation system results are signified by two ways as: (i) Comparative analysis in the form of reduced bad or under segmentation rate with the existing Modi character segmentation approach presented in [22]. It is given in Table 2. (ii) Comparative analysis with benchmarking state-of-art character segmentation approaches. So, character segmentation benchmarking state-of-art approaches are studied and implemented. The overall comparative analysis is given in

Table 2 Comparative analysis with existing Modi character segmentation approach presented in [22]

Approach	NC_R	NC_C	NC_B	NC_I
Existing approach [Deshmukh M.S. & Kolhe S.R. 2019][22]	28,983	4094	238	24,651
Modified proposed approach	**29,069**	**3495**	**80**	**25,494**

The results of proposed approach are in bold

Table 3 Comparative analysis with state-of-art character segmentation approaches

Approach	NC_R	NC_C	NC_B	NC_I	SSR (%)	CSR (%)	Precision	Recall	F-measure
[23]	35,748	4345	9330	12,073	33.77	66.26	0.5	0.5	0.5
[24]	36,011	6010	16,004	13,997	38.87	42.02	0.6	0.6	0.6
[25]	34,081	4806	7784	21,491	63.06	64.51	0.8	0.8	0.8
[29]	32,395	6545	5534	20,316	62.71	60.98	0.8	0.8	0.8
[27]	26,894	1832	6531	18,531	68.90	55.63	0.8	0.6	0.7
Proposed approach	**29,069**	**3495**	**80**	**25,494**	**87.70**	**76.53**	**1.0**	**0.9**	**0.9**

The results of proposed approach are in bold

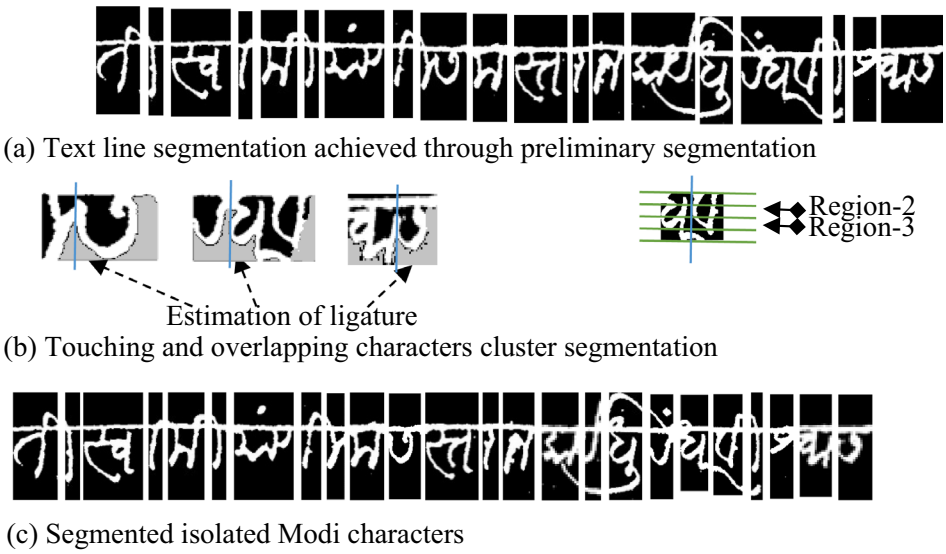


Fig. 7 Illustration of Modi character segmentation with overlapping/ touching character cluster segmentation

(a) Text line segmentation achieved through preliminary segmentation

(b) Touching and overlapping characters cluster segmentation

(c) Segmented isolated Modi characters



Fig. 8 Modi character segmentation (overlapping and touching characters)

(a) Input Modi text line

(b) Segmented Modi characters

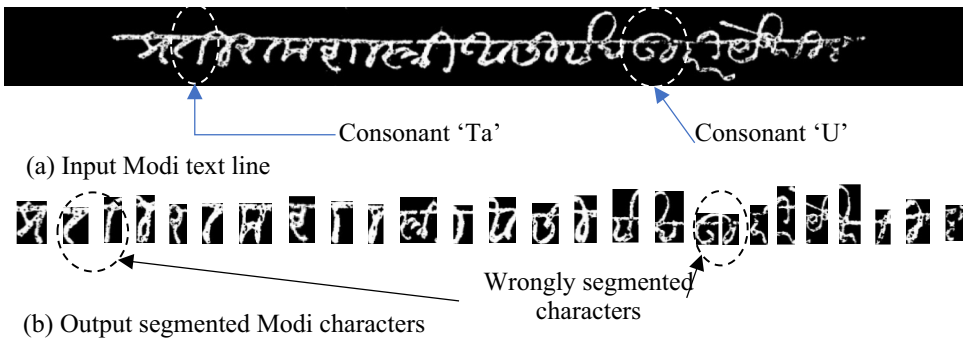


Fig. 9 Example of Modi script character segmentation (overlapping, non-uniform Shirorekha and baseline, variable size height and width of characters)

(a) Input Modi text line

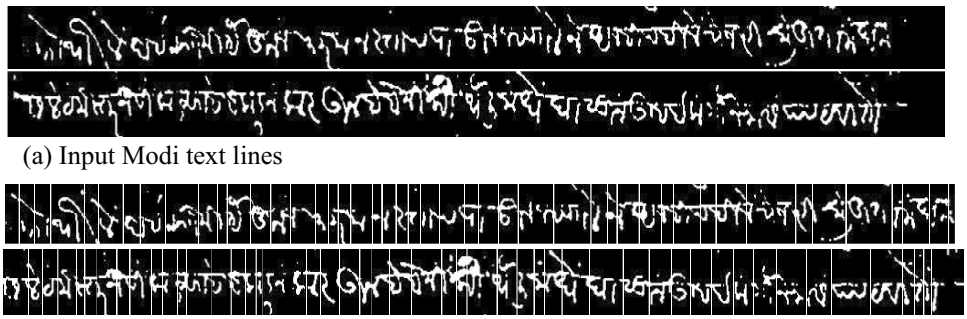
(b) Output segmented Modi characters



(a) Input Modi text line

(b) Segmented Modi characters

Fig. 10 Modi character segmentation (variable distance in characters)



(a) Input Modi text lines

(b) Segmented Modi characters

Fig. 11 Modi character segmentation (broken characters and Shirorekha, non-uniform base line and Shirorekha)



(a)Poorly segmented Modi characters cluster

(b)Partially segmented Modi characters cluster



(c) Un-segmented Modi characters cluster

Fig. 12 Example of unsegmented Modi script character clusters

Table 3. Efficiency of each approach is evaluated using the same Modi dataset.

Two hundred and thirty eight bad characters are segmented by the existing Modi character segmentation technique presented in [22]. This is reduced to eighty bad characters by the proposed approach. Means, as compare to previous Modi character segmentation technique, proposed modified technique reduced bad or under segmentation rate from 0.8% to 0.2% with increase in Correct Segmentation Rate inexact to 2%. Over segmentation rate is also reduced. Correct Segmentation Rate (CSR) is achieved up to 76.53%. Instead of ground truth values, if numbers of estimated characters in text lines are determined then it gives Successful Segmentation Rate (SSR) as 87.70% with precision 1.0, Recall and F-measure as 0.9. Figures 7, 8, 9, 10, 11, 12, 13, 14 shows examples of Modi character segmentation which clarifies the efficiency of the proposed approach.

Character segmentation approach presented in [23] is efficiently segmented isolated Modi characters. This approach is failed to segment overlapping and touching characters. Text line is not segmented when baseline and top line is non-uniform/curved. Character segmentation approach proposed by [24] is failed to segment broken characters and cursive Modi text lines. This approach segments only well written text lines. Character segmentation approach discussed



(a) Segmented text lines of a Modi document

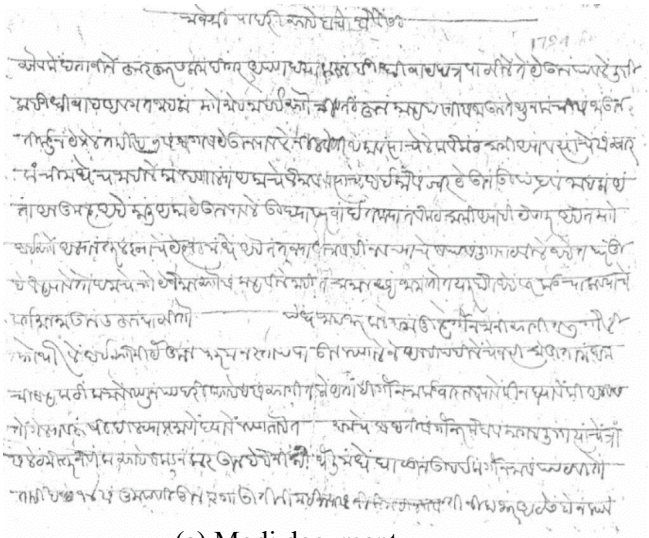
(b) Segmented characters of text lines

Fig. 13 Example of Modi script character segmentation of a Modi document text lines

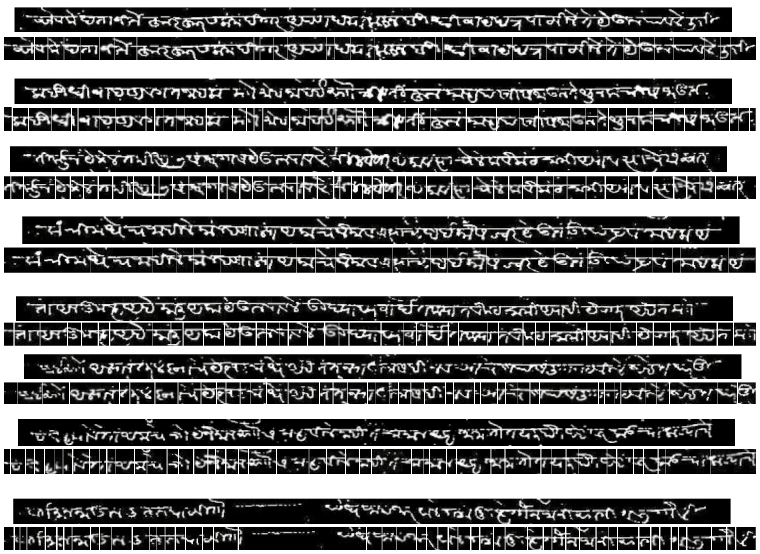
in [25] and [29] efficiently segments Modi text lines into isolated characters and clusters of overlapping/touching

characters. These approaches are not segment overlapping/touching character clusters. Location of Shirorekha of Modi text line is efficiently detected by the method proposed in [27] and text region is efficiently identified. This method is not segment overlapping and touching characters. All these methods are exceedingly having the problem of bad or under segmentation.

Modi script is adapted for fast writing purpose in administration. Consequently, text line includes overlapping and touching characters. Previously proposed character segmentation approach in [22] is handling overlapping and touching characters challenge precisely. In Modi consonant 'Ta' and vowel 'U' are written as shown in Fig. 9a. These Modi characters looks like two separate characters when it is written as individually and hence segmented poorly as shown in Fig. 9b. In Fig. 10 and Fig. 13, even though the distance between two characters is inconsistent Modi characters are segmented properly. Likewise, Fig. 11 illustrates novelty of the proposed approach related to efficiently handling of three character segmentation challenges as: (i) broken characters, (ii) non-uniform and broken Shirorekha and (iii) non-uniform baseline. Very tightly coupled and highly degraded touching or overlapping characters in clusters look like a single character. Hence, they are not segmented properly for example as shown in Fig. 12c. Similarly, Fig. 12a and b shows the example of poorly segmented and partially segmented Modi character clusters. Hence, there is a need to work on the segmentation of very tightly and highly degraded overlapping/touching Modi characters clusters. Efficiency of the proposed approach of handling all the challenges given in Table 1 is illustrated by the example shown in Fig. 14.



(a) Modi document



(b) Some segmented text lines and segmented characters of a Modi document in (a)

Fig. 14 Example of Modi script character segmentation of a Modi document

Conclusion

Efficiency of intelligent handwritten ancient document recognition, transcription/transliteration or digitization system is highly depends on the accuracy of character segmentation. Hence, in these processes character segmentation is the vital process. A robust, modified character segmentation approach is proposed for handwritten archaic Modi script documents. This character segmentation approach is based on local and global zoning with background and foreground pixel intensity analysis. Total 87.70% Modi character segmentation rate is achieved with 0.2% segmentation error rate. The result proves efficiency of the proposed modified preliminary stage. It is feasible and effective to segment isolated and overlapping/touching Modi character clusters. It reduces bad segmentation rate up to adequate state as compare to existing hybrid background pixel density based character segmentation technique. Consequently, under segmentation problem with loss of information is eliminated. The proposed Modi document character segmentation approach is expertly dealing with the non-uniform Shirorekha and baseline. Proposed approach segments the characters by scrutinizing the segmentation column two times and then finalised the segmentation region of the character. Hence, the presented approach is efficiently tackled with the character segmentation challenges like broken character, broken Shirorekha, variable size characters, variable distance between characters, skewed text lines and curved text lines. To segment overlapping/touching characters efficiently it is necessary to identify overlapping region of the characters cluster. However, further to improve the accuracy of Modi character segmentation, there is a need to combine all useful information from handwriting and overcome all the overlapping/touching characters cluster segmentation challenges.

Data availability The datasets used are available on the IEEE Dataport and Kaggle datasets repository.

Declarations

Conflict of interest The authors have no competing interests.

References

1. Choudhary A, Rishi R, Ahlawat S. A new character segmentation approach for off-line cursive handwritten words. Procedia Comput Sci. 2013;17:88–95.
2. Kumar M, Jindal MK, Sharma RK. Segmentation of isolated and touching characters in offline handwritten Gurmukhi script recognition. Int J Inform Technol Comput Sci. 2014;6(2):58–63.
3. Kurniawan F, Rahim MSM, Daman D, Rehman A, Mohamad D, Mariyam S. Region-based touched character segmentation in handwritten words. Int J Innov Comput Inf Control. 2011;7(6):3107–20.
4. Garg NK, Kaur L, Jindal MK. Segmentation of touching modifiers and consonants in middle region of handwritten Hindi text. Pattern Recognit Image Anal. 2015;25(3):413–7.
5. Saba T, Rehman A, Elarbi-Boudihir M. Methods and strategies on off-line cursive touched characters segmentation: a directional review. Artif Intell Rev. 2014;42(4):1047–66.
6. Sharma P, Sachan MK. A review on character segmentation of touching and half character in handwritten hindi text. Int J Adv Res Comput Sci. 2017;8(3):1078.
7. Jindal K, Kumar R. A novel shape-based character segmentation method for devanagari script. Arab J Sci Eng. 2017;42(8):3221–8.
8. Bhowmik TK, Roy A, Roy U. Character segmentation for handwritten Bangla words using artificial neural network. Proc. 1st IAPR TC3 NNLDAR; 2005.
9. Basu S., Chaudhuri C., Kundu M., Nasipuri M., & Basu D. K. (2012). Segmentation of offline handwritten bengali script. arXiv preprint arXiv:1202.3046.
10. Roy A, Bhowmik TK, Parui SK, Roy U. A novel approach to skew detection and character segmentation for handwritten Bangla words. In: Digital image computing: techniques and applications, 2005 .DICTA'05. Proceedings 2005. IEEE; 2005. p. 30–30.
11. Palakollu S, Dhir R, Rani R. Handwritten Hindi text segmentation techniques for lines and characters. In: Proceedings of the world congress on engineering and computer science, vol I, WCECS 2012, San Francisco, USA, 24–26 October 2012.
12. Bag S, Krishna A. Character segmentation of hindi unconstrained handwritten words. In: International workshop on combinatorial image analysis. Cham: Springer; 2015. p. 247–60.
13. Golait SS, Malik L. Handwritten marathi compound character segmentation using minutiae detection algorithm. Procedia Comput Sci. 2016;87:18–24.
14. Kapoor S, Verma V. Fragmentation of handwritten touching characters in devanagari script. Int J Inform Technol Model Comput (IJITMC). 2014;2:11–21.
15. Deshmukh MS, Patil MP, Kolhe SR. The divide-and-conquer based algorithm to detect and correct the skew angle in the old age historical handwritten modi lipi documents. Int J Comput Sci Appl. 2017;14(2):47–63.
16. Deshmukh MS, Patil MP, Kolhe SR. A dynamic statistical non-parametric cleaning and enhancement system for highly degraded ancient handwritten modi lipi documents. In: Advances in computing, communications and informatics (ICACCI), 2017 international conference on. IEEE; 2017. p. 1545–51.
17. Behera S, Pradhan A, Majhi B. A novel clustering based fuzzy approach for character segmentation in handwritten odia scripts. In: Image information processing (ICIIP), 2017 fourth international conference on. IEEE; 2017. p. 1–6.
18. Kavitha AS, Shivakumara P, Kumar GH, Lu T. A new watershed model based system for character segmentation in degraded text lines. AEU-Int J Electron Commun. 2017;71:45–52.
19. Peng G, Yu P, Li H, Li H, Zhu X. A character segmentation algorithm for the palm leaf manuscripts. In: Computational intelligence and applications (ICCIA), 2017 2nd IEEE international conference on. IEEE; 2017. p. 354–8.
20. Kavallieratou E, Stamatatos E, Fakotakis N, Kokkinakis G. Handwritten character segmentation using transformation-based learning. In: Icpr. IEEE; 2000. p. 2634.
21. Deshmukh MS, Patil MP, Kolhe SR. A hybrid text line segmentation approach for the ancient handwritten unconstrained freestyle Modi script documents. Imaging Sci J. 2018;66(7):433–42.
22. Deshmukh M. S. & Kolhe S. R. (2019). A Hybrid Character Segmentation Approach for Cursive Unconstrained Handwritten Historical Modi Script Documents. International conference

on sustainable computing in science, technology & management (SUSCOM-2019) 26th -28th Feb 2019, SSRN Elsevier Digital Library
23. Narang SR, Jindal MK, Kumar M. Drop flow method: an iterative algorithm for complete segmentation of Devanagari ancient manuscripts. Multimedia Tools Appl. 2019;78:23255–80.
24. Sahare P, Dhok SB. Robust character segmentation and recognition schemes for multilingual Indian document Images. IETE Tech Rev. 2019;36(2):209–22.
25. Singh AP, Kushwaha AK. Analysis of segmentation methods for brahmi script. DESIDOC J Library Inform Technol. 2019;39(2):109–16.
26. Ahmad I, Wang X, Li R, Ahmed M, Ullah R. Line and ligature segmentation of urdu nastaleeq text. IEEE Access. 2017;5:10924–40.
27. Roy PP, Bhunia AK, Das A, Dey P, Pal U. HMM-based Indic handwritten word recognition using zone segmentation. Pattern Recogn. 2016;60:1057–75.
28. Borah S. Segmentation of assamese handwritten characters based on projection profiles. Int J Comput Appl. 2015;130(17):12–5.
29. Kosarat R, Hiransakolwong N. Segmentation of touching character printed lanna script using junction point. J Eng Sci Technol. 2018;13(10):3331–43.
30. Murthy OV, Hanmndlu M. An approach to offline handwritten Devanagari word segmentation. Int J Comput Appl Technol. 2012;44(4):284–92.
31. Deshmukh MS, Kolhe SR. Unsupervised page area detection approach for the unconstrained chronic handwritten modi document images. In: 2021 international conference on emerging smart computing and informatics (ESCI). IEEE; 2021. p. 130–5.
32. Deshmukh Manisha, Kolhe Satish, 2023 “MODI-HHDoc: historical MODI script handwritten document dataset”, Mendeley Data, V1, https://doi.org/10.17632/sg337vf6wn.1.
33. Manisha Deshmukh, Satish Kolhe, June 6, 2023, "MODI-HHDoc: historical MODI script handwritten document dataset". IEEE Dataport. https://doi.org/10.21227/1z10-w986.

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